Impact of Social Media on Mental Health

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Abstract

Excessive social media use has been increasingly associated with adverse mental health outcomes. However, most of the existing studies have an emphasis on only predictive academic studies and provide limited practical guidance and help for psychological intervention. The paper aims to build a data-driven system that models social media behavior as a structured and interpretable behavioral profile, integrating predictive modeling with retrieval-based recommendation support. Using the Time-Wasters on Social Media dataset, user behavior is categorized into four complementary dimensions: usage intensity, interaction habits, content preference, and self-regulation. These behavioral signals are aggregated into interpretable features for downstream modeling. The resulting profiles are employed for risk prediction and are converted into semantic queries to retrieve relevant guidance from expert-informed resources.

System evaluation comprised both predictive and analytical experiments. The binary classification achieved moderate performance, with a 0.63 area under the curve, indicating that behavioral features capture meaningful, though not fully separable, risk patterns. Additional analysis applied a three-level risk threshold for grouped ablation that reveals the content and motivation features may introduce noise to the multiclass setting. An interactive model is built as the output of the paper to provide practical guidance for self-test and pre-psychological consulting uses.

Keywords: social media mental health, behavioral profiling, addiction risk prediction, retrieval-augmented guidance, ablation study, self-regulation, user behavior modeling, multiclass risk classification, recommendation system.

1 Introduction

1.1 Social Problem

Social media platforms have become deeply embedded in daily life, particularly among adolescents and young adults. While they facilitate communication and information sharing, excessive usage has been associated with negative mental health outcomes. For example, a study in Bangladesh found that more than four hours of daily use is linked to higher risks of anxiety and depression among young adults (4). With the increasing prevalence of smartphones and algorithm-driven content delivery, this issue requires growing attention.

Although social media use is widely recognized as a general risk factor, it is critical to identify which specific behavioral patterns contribute most to harmful outcomes. Social media engagement is multifaceted, involving not only total usage time but also interaction frequency and content consumption patterns. Therefore, understanding these behavioral dimensions is essential for effective intervention.

1.2 Technical Solution

Existing studies often focus on predictive modeling but provide limited practical guidance. This project addresses this gap by developing a system that produces interpretable risk assessments alongside actionable recommendations.

We construct a data-driven framework that transforms raw usage data into structured behavioral profiles. These profiles are used to analyze user patterns and generate both risk predictions and guidance for healthier online habits. Raw features such as screen time, session frequency, and interaction metrics are normalized and aggregated to capture both short-term and long-term behaviors.

User behavior is represented across four dimensions: usage intensity, interaction patterns, content preference, and self-regulation. To capture

higher-level patterns, these features are further aggregated into summary scores (usage score, interaction score, and self-regulation risk), which serve as inputs to a binary classifier for risk prediction. Risk labels are defined based on outcomes such as addictive engagement and productivity loss.

Tree-based models (e.g., XGBoost) are employed to handle structured and mixed-type features, while class imbalance is addressed through weighted training and precision-recall analysis. Ablation studies are conducted to quantify the contribution of each behavioral dimension.

Finally, an interactive system allows users to input their behavior, construct a profile, and retrieve evidence-based guidance from expert sources such as WHO and SAMHSA, enabling both risk assessment and personalized recommendations.

The goal of this project is to develop an interpretable system that predicts social media addiction risk and provides personalized, retrieval-based guidance for early intervention.

1.3 Relationship between Social Media Usage and Mental Health

Prior work on social media and mental health can be grouped into several themes.

1.3.1 Usage Time Focused Studies

Riehm et al. find that individuals spending more than three hours per day on social media exhibit significantly higher rates of anxiety and depression, suggesting that prolonged exposure is a key risk factor beyond mere access or frequency (2).

1.3.2 Addiction-focused Behavioral Analyses

Haweny et al. show that social media addiction, rather than raw usage time, is associated with reduced academic productivity, emphasizing psychological mechanisms such as ego depletion and delayed gratification (5). Similarly, Griffiths proposes a multi-component model of addiction, arguing that behavioral addictions arise from interacting factors (e.g., salience, withdrawal, relapse), highlighting the need to model addiction as a multi-dimensional construct rather than through single indicators (6).

1.3.3 Computational Modeling of Social Media Behavior

Chawla et al. introduce SMOTE to address class imbalance, demonstrating improved performance in ROC space and motivating the use of AUC-based evaluation (3). These techniques are particularly relevant for modeling imbalanced behavioral risk categories in our setting.

1.3.4 Behavioral Patterns and Health Outcomes

Paul and Headley-Johnson show that social media effects depend not only on usage intensity but also on content type and interaction patterns. In particular, passive consumption (e.g., scrolling) is associated with worse outcomes than active engagement. These findings motivate the use of multi-dimensional behavioral features and support the translation of behavioral patterns into actionable recommendations in our system (7).

1.3.5 Ablation Studies

Kerz et al. demonstrate that feature ablation is effective for identifying informative feature groups in mental health prediction tasks. This motivates our use of grouped ablation to quantify the contribution of behavioral dimensions in multiclass risk prediction (1).

2 Methods

2.1 Method 1: Behavioral Feature Engineering / Behavior Profile Construction

We construct a structured behavioral profile for each user to capture the multi-dimensional nature of social media risk and to enable both prediction and retrieval. Raw tabular data are transformed into a semantically meaningful representation that supports downstream modeling and embedding-based retrieval.

User behavior is organized into four dimensions: usage intensity (time spent, sessions, videos watched), interaction patterns (scrolling rate, engagement), content preference (video categories, watch reasons), and self-regulation (self-control, satisfaction). Irrelevant variables (e.g., time-of-day frequency and identifiers) are excluded. Numerical features are standardized via z-score normalization, and categorical variables are one-hot encoded.

To capture higher-level patterns, features are aggregated into three summary scores: usage score, interaction score, and self-regulation risk. The self-regulation score is negated to ensure consistent risk directionality. Selected raw variables (e.g., scrolling rate and self-control) are retained to preserve fine-grained signals.

For retrieval, profiles are converted into natural language queries (e.g., “high screen time, frequent

scrolling, low self-control") and matched to expert resources (WHO, SAMHSA). Compared to symbolic tuples, this representation is more expressive and better suited for semantic search. Self-regulation combines self-control and satisfaction to provide a more robust indicator of behavioral risk. Demographic variables are excluded to improve generalizability.

The final dataset represents each user as a unified profile of aggregated scores, selected features, one-hot encoded content variables, and target labels (addiction level, productivity loss). This representation serves both prediction and retrieval, where features are mapped to qualitative descriptors for query generation.

Sanity checks indicate multi-dimensional effects, with self-regulation providing the strongest signal.

2.2 Method 2: Gradient Boosted Decision Framework for Risk Classification

Recommendations are triggered based on both predicted risk and retrieval relevance.

Addiction_Level is selected as the basis for risk labeling because it directly measures the degree of addictive engagement as an outcome, rather than serving as a behavioral or dispositional predictor. Other candidate variables in the dataset either reflect behavioral patterns during consumption or individual-level traits, neither of which directly captures the extent of harm experienced by the user. In contrast, *Addiction_Level* captures the actual degree of addictive behavior, making it the most direct measure of risk outcome. The validity of this choice is further supported by its strong association with *Self_Control* ($r = -0.84$) and *Productivity-Loss* ($r = -0.83$), suggesting that *Addiction_Level* reflects a broader harm outcome that converges across multiple behavioral dimensions.

Based on the behavioral profiles constructed in Section 2.1, the input to the model is a user profile vector comprising behavioral scores and categorical indicators. The target variable is *risk_binary*, derived from addiction level as a binary grouping of low-risk versus high-risk users. *Addiction_Level* is excluded from the feature set to prevent label leakage, as *risk_binary* is directly derived from it.

We then adopt XGBoost (eXtreme Gradient Boosting) as the classification backbone to predict binary addiction risk from user behavioral profiles. XGBoost is selected over deep neural networks for three reasons: the dataset consists of structured, human-engineered behavioral features rather than raw signals; the sample size is moderate, where tree-based ensembles consistently outperform neural networks; and the mixed feature space (continuous behavioral scores and one-hot encoded categorical variables) requires no additional preprocessing under a tree-based framework.

The binary risk classifier achieves an AUC of 0.63. We optimize the decision threshold to favor recall over precision, since missing a high-risk user is more costly than occasionally forwarding a low-risk user to retrieval. At the selected threshold of 0.48, the model identifies 82 % of high-risk users correctly.

The moderate overall performance is consistent with the behavioral analysis conducted in Section 2.1. As established in our sanity checks, usage and interaction features (e.g., *Scroll_Rate*, *usage_score*) exhibit minimal separation between risk groups, while self-regulation variables show the strongest discriminative signal. Since self-regulation features are excluded from the classifier to prevent label leakage; the remaining behavioral and content features provide a weaker but unbiased signal for risk prediction. The result shows that *Watch_Reason_Boredom* emerges as the most influential feature, where the absence of boredom-driven consumption is associated with higher predicted risk.

2.3 Method 3: Retrieval

Our system adopts a retrieval-based approach to transform user behavior into actionable guidance. Rather than relying solely on prediction, it interprets behavioral patterns, aligns them with expert knowledge, and returns explanations and recommendations.

Based on the behavioral profiles constructed in Method 1, each user is converted into a semantic query (e.g., "high screen time, frequent scrolling, low self-control"), enabling compatibility with natural language retrieval. The query is matched against a corpus of expert documents from WHO and SAMHSA. Documents are segmented into smaller chunks and encoded into dense vector representations, and cosine similarity is used to measure relevance between the query and each chunk.

The system retrieves the top-K most relevant segments (typically 3–5).

To ensure recommendation quality, two thresholds are applied: a minimum *Addiction_Level* requirement to trigger recommendations, and a min-

imum similarity score to filter out low-relevance matches.

The selected segments are then used to generate user-facing outputs, including explanations and recommendations. Outputs may be presented as direct quotes or summarized text, allowing behavioral patterns to map dynamically to evidence-based guidance.

2.4 Method 4: Analysis Through Ablation Studies

2.4.1 Analysis Setup

To evaluate the system's ability to identify risky behaviors, we analyze representative cases and examine which behavioral signals contribute most to model outputs. Instead of a binary label, we define a three-level risk variable based on *Addiction_Level*: low (0–2), medium (3–4), and high (5–7). The dataset contains 488 low-risk, 195 medium-risk, and 317 high-risk users, resulting in class imbalance.

A leakage check is performed prior to modeling. Strong inverse correlations are observed between *Addiction_Level* and *Self_Control* ($r = -1.000$) as well as *ProductivityLoss* ($r = -0.995$). These proxy variables are therefore removed to ensure the analysis captures meaningful behavioral patterns rather than trivial label reconstruction.

2.4.2 Ablation Study

We conduct a grouped ablation study to evaluate feature importance. Starting from the full model, we iteratively remove one feature group, retrain under the same train-test split, and compare changes in macro-F1.

The feature groups include: summary behavior (`usage_score`), interaction and self-regulation (`interaction_score`, `self_reg_risk`, `Scroll_Rate`), content preference (`Video_Category_*`), and motivation (`Watch_Reason_*`).

The full model achieves an accuracy of 0.355 and a macro-F1 of 0.342. Removing interaction and self-regulation features reduces macro-F1 by approximately 3%, indicating moderate predictive contribution. In contrast, removing motivation and content preference features increases macro-F1 by 10.8% and 15.8%, respectively, suggesting that these features introduce noise or redundancy in the current model.

These findings suggest that the most important signals in the current pipeline are not content-based features but interaction intensity patterns. It is a meaningful conclusion as it shows that the behavioral risk is better captured through how users engage with social media than through what content categories they consume.

3 Result

We evaluate mental health risk prediction as a binary classification task, where users with high addiction or productivity loss are labeled as high-risk. The objective is to accurately identify high-risk users based on behavioral profiles, while supporting downstream interactive recommendations.

The dataset is split into training, validation, and test sets using stratified sampling. Model performance is evaluated using precision, recall, F1-score, and ROC-AUC, with particular emphasis on recall to minimize missed high-risk users.

We adopt XGBoost due to its effectiveness on structured behavioral data. The decision threshold is optimized under a recall constraint (High Risk recall ≥ 0.90), yielding a threshold of 0.4871 with recall = 0.90 and precision = 0.54.

The model achieves a cross-validation AUC of 0.5729 ± 0.0256 and a test AUC of 0.5811. At the default threshold, the macro-F1 is 0.56 (Low Risk: 0.53, High Risk: 0.60). Under the optimized threshold, High Risk recall increases to 0.90 while precision decreases to 0.54, reflecting a deliberate trade-off favoring sensitivity over specificity. This is further illustrated by the imbalance between High Risk recall (0.90) and Low Risk recall (0.19), indicating that the model prioritizes minimizing false negatives at the cost of increased false positives.

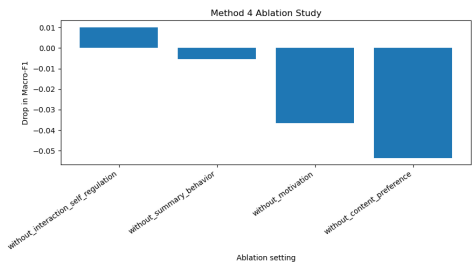

Figure 1: Ablation Study

For multiclass risk prediction, behavioral features provide moderate separability across low-, medium-, and high-risk users. As shown in Figure 1, removing interaction and self-regulation features slightly decreases macro-F1, while removing motivation and content features increases macro-F1

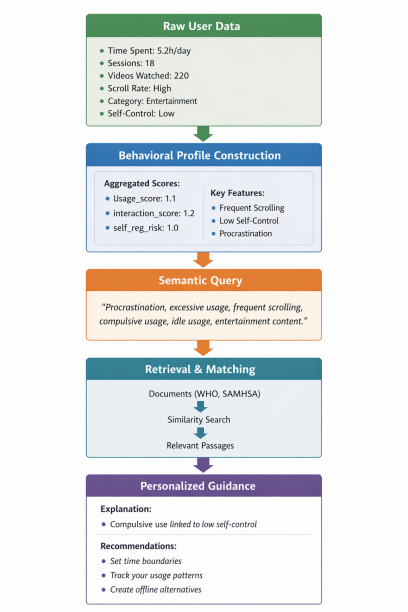

Figure 2: Simplified Flowchart Example of Process

by 10.7% and 15.7%, indicating that these features introduce noise in the current model.

As shown in Figure 2, the retrieval component maps behavioral profiles into natural language queries and retrieves relevant guidance using SentenceTransformer embeddings and cosine similarity. Queries from high-risk profiles (e.g., "high usage," "low self-control," "procrastination") consistently return semantically aligned content on addiction and self-regulation strategies, demonstrating effective semantic matching beyond keyword overlap.

4 Conclusion

This project addresses the impact of excessive social media use on mental health by developing a data-driven system that models user behavior as multi-dimensional profiles (usage intensity, interaction patterns, content preference, and self-regulation). By transforming raw data into structured behavioral representations, the system supports both risk prediction and retrieval-based guidance.

The main contribution lies in integrating behavioral modeling with semantic retrieval. We (1) construct interpretable behavioral profiles, (2) develop a classifier for risk prediction, and (3) design a retrieval pipeline that maps behavioral profiles to expert guidance (e.g., WHO, SAMHSA). This unified framework enables both identification of high-risk users and delivery of personalized recommendations.

Experimental results show that behavioral risk is driven by interactions among multiple dimensions rather than any single feature. Self-regulation features provide the strongest signal for distinguishing high-risk users, while content-based features contribute less and may introduce noise. The classification model achieves moderate performance (AUC ≈ 0.63), indicating that behavioral features alone provide limited separability.

Several limitations remain. Model performance is constrained by weakly separable patterns and potential label ambiguity in the dataset. In addition, retrieval quality depends on similarity thresholds and document availability, limiting the strength of recommendations for some users.

Future work includes incorporating richer data (e.g., temporal behavior or textual content), improving retrieval with advanced embedding or reranking methods, and integrating user feedback for adaptive recommendations. Extending the system toward a more interactive or agent-based framework may further enhance its practical impact.

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